

TUSHAR SONI

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EDUCATION

Jaypee University of Engineering and Technology

Guna, Madhya Pradesh, India

Bachelor of Technology in Computer Science and Engineering (2027 Batch)

Aug 2023 – May 2027

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming, Software Development Life Cycle (SDLC)

LANGUAGES & TECHNOLOGIES

Languages: JavaScript (ES6+), TypeScript, SQL, C++

Frontend: React.js, REST API Integration, WebSockets (Socket.io)

Backend: Node.js, Express.js, REST API Design, JWT, OAuth2, BullMQ

Databases: MongoDB, MySQL, PostgreSQL, Redis, Drizzle ORM, Mongoose ODM

Tools & Practices: Git, GitHub, Docker, Postman, Linux, Version Control, Code Documentation

PROJECTS

JettMessage – Real-Time E2EE Chat Application | *React.js, Node.js, PostgreSQL, MongoDB, Redis, BullMQ* | [GitHub](#)

- Developed a full-stack, real-time web application using React.js on the frontend and Node.js/Express.js on the backend, following frontend/backend separation of concerns throughout the SDLC.
- Designed and integrated REST APIs for user authentication, session management, and message delivery, consumed by the React.js frontend using standard HTTP and WebSocket protocols.
- Built a zero-knowledge messaging architecture with client-side asymmetric key generation (IndexedDB) and double-encrypted payload routing, ensuring the server never stores plaintext content.
- Implemented real-time presence and reliable message delivery using WebSockets (Socket.io) and Redis state management, with BullMQ handling offline message queuing across reconnections.
- Modelled multi-participant conversation state and user identities in PostgreSQL using Drizzle ORM; maintained clear code documentation and Git commit history throughout development.

GateX – Scalable API Gateway | *Node.js, Express.js, PostgreSQL, Redis, Docker, Git* | [GitHub](#)

- Built and deployed a production-grade, scalable web application in Node.js to handle request routing, cross-service orchestration, and centralized authentication across multi-service backend architectures.
- Wrote clean, maintainable JavaScript following REST API design best practices; troubleshooted and debugged routing and proxy issues throughout the development lifecycle.
- Reduced per-request synchronous database overhead by **50ms** by designing an asynchronous logging pipeline with a Redis queue and a scheduled background worker bulk-inserting metrics into PostgreSQL.
- Engineered a sliding-window rate limiter using Redis Sorted Sets, eliminating fixed-window boundary exploits and protecting downstream services from traffic spikes.
- Containerized the full service with Docker for reproducible deployments; managed all source control via Git with documented version history.

COMPETITIVE PROGRAMMING

LeetCode

2023 – Present

- Solved **150+** problems in C++ across arrays, linked lists, trees, strengthening core problem-solving and analytical skills.

INTERESTS

Full-Stack Web Development, Backend System Design, REST API Development, Open Source Contribution, DevOps